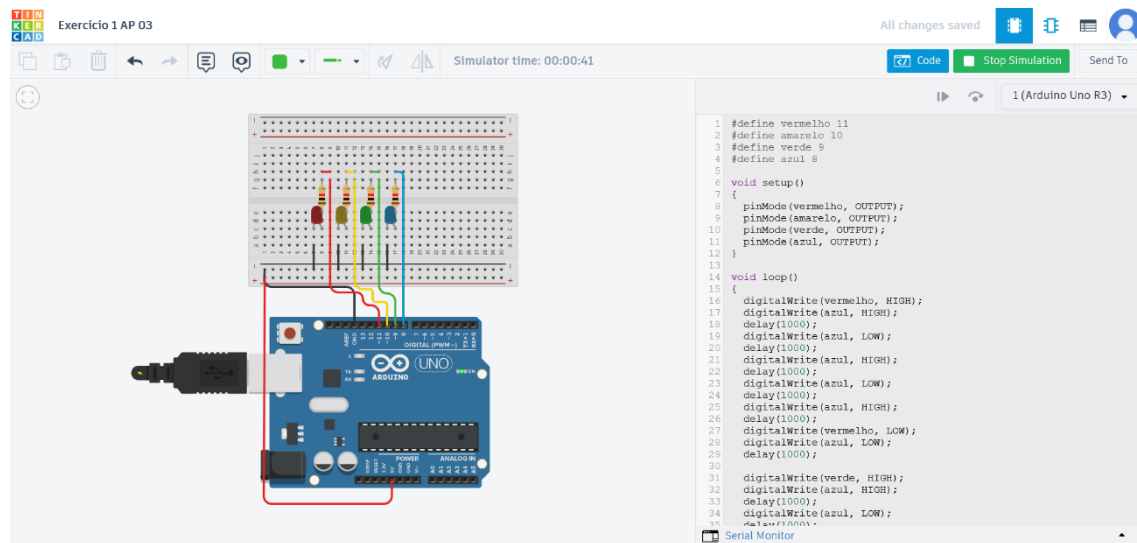


LABORATÓRIO DE AC

GABRIEL EVANGELISTA MASSARA

EXERCÍCIO PRÁTICO 03 – PARTE 1

PRINTS DO TINKERCAD ELABORADO:



CÓDIGO:

```
1 #define vermelho 11
2 #define amarelo 10
3 #define verde 9
4 #define azul 8
5
6 void setup()
7 {
8   pinMode(vermelho, OUTPUT);
9   pinMode(amarelo, OUTPUT);
10  pinMode(verde, OUTPUT);
11  pinMode(azul, OUTPUT);
12 }
13
14 void loop()
15 {
16   digitalWrite(vermelho, HIGH);
17   digitalWrite(azul, HIGH);
18   delay(1000);
19   digitalWrite(azul, LOW);
20   delay(1000);
21   digitalWrite(azul, HIGH);
22   delay(1000);
23   digitalWrite(azul, LOW);
24   delay(1000);
25   digitalWrite(azul, HIGH);
26   delay(1000);
27   digitalWrite(vermelho, LOW);
28   digitalWrite(azul, LOW);
29   delay(1000);
30
31   digitalWrite(verde, HIGH);
32   digitalWrite(azul, HIGH);
33   delay(1000);
34   digitalWrite(azul, LOW);
```

```

35 delay(1000);
36 digitalWrite(azul, HIGH);
37 delay(1000);
38 digitalWrite(azul, LOW);
39 delay(1000);
40 digitalWrite(azul, HIGH);
41 delay(1000);
42 digitalWrite(azul, LOW);
43 delay(1000);
44 digitalWrite(azul, HIGH);
45 delay(1000);
46 digitalWrite(azul, LOW);
47 digitalWrite(verde, LOW);
48 delay(1000);
49
50 digitalWrite(amarelo, HIGH);
51 digitalWrite(azul, HIGH);
52 delay(1000);
53 digitalWrite(azul, LOW);
54 delay(1000);
55 digitalWrite(azul, HIGH);
56 delay(1000);
57 digitalWrite(azul, LOW);
58 digitalWrite(amarelo, LOW);
59 delay(1000);
60 }

```

EXERCÍCIO PRÁTICO 03 – PARTE 2

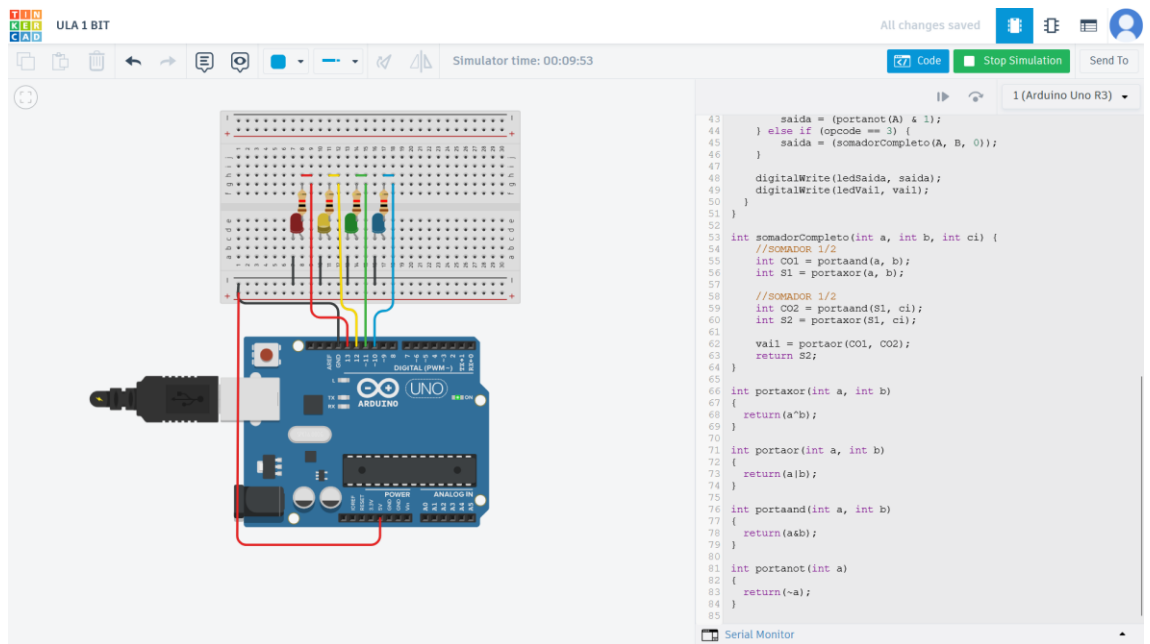
1. TABELA COM AS INSTRUÇÕES

Instrução realizada	Binário (A,B,Op.code)	Valor em Hexa (0x ...)	Resultado em binário
AND(A,B)	0 1 00	0x4	0
OR(A,B)	1 0 01	0x9	1
SOMA(A,B)	1 0 11	0xB	1
NOT(A)	0 0 10	0x2	1
AND(B,A)	0 1 00	0x4	0

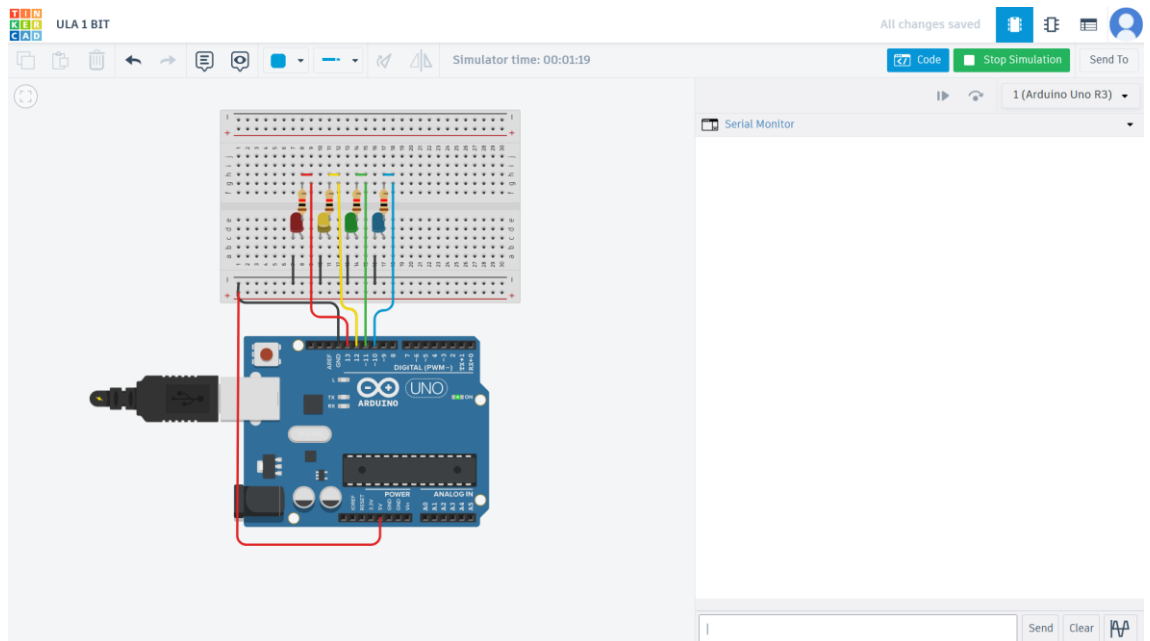
2. PRINTS DA MONTAGEM:

```
1  #define vermelho 13
2  #define amarelo 12
3  #define ledSaida 11
4  #define ledVail 10
5
6  int A = 0;
7  int B = 0;
8  int opcode = 0;
9
10 int saida = 0;
11 int vail = 0;
12
13 void setup()
14 {
15     Serial.begin(9600);
16     pinMode(vermelho, OUTPUT);
17     pinMode(amarelo, OUTPUT);
18     pinMode(ledSaida, OUTPUT);
19     pinMode(ledVail, OUTPUT);
20 }
21
22 void loop()
23 {
24     if(Serial.available()) {
25         String entrada = Serial.readStringUntil('\n');
26         entrada.trim();
27
28         A = entrada.charAt(0) - '0';
29         B = entrada.charAt(1) - '0';
30         opcode = entrada.charAt(2) - '0';
31
32
33         digitalWrite(vermelho, A);
34         digitalWrite(amarelo, B);
35
36         //AND
37         if(opcode == 0) {
38             saida = portaand(A, B);
39         } else if (opcode == 1) {
40             saida = portaor(A, B);
41         } else if (opcode == 2) {
42             //Uso esse &1 para converter para inteiro,
43             saida = (portanot(A) & 1);
```

```
44     } else if (opcode == 3) {
45         saida = (somadorCompleto(A, B, 0));
46     }
47
48     digitalWrite(ledSaida, saida);
49     digitalWrite(ledVai1, vai1);
50 }
51 }
52
53 int somadorCompleto(int a, int b, int ci) {
54     //SOMADOR 1/2
55     int CO1 = portaand(a, b);
56     int S1 = portaxor(a, b);
57
58     //SOMADOR 1/2
59     int CO2 = portaand(S1, ci);
60     int S2 = portaxor(S1, ci);
61
62     vai1 = portaor(CO1, CO2);
63     return S2;
64 }
65
66 int portaxor(int a, int b)
67 {
68     return (a^b);
69 }
70
71 int portaor(int a, int b)
72 {
73     return (a|b);
74 }
75
76 int portaand(int a, int b)
77 {
78     return (a&b);
79 }
80
81 int portanot(int a)
82 {
83     return (~a);
84 }
85
```



3. EXECUÇÃO



ULA 1 BIT

All changes saved

Simulator time: 00:02:23

CodeStop SimulationSend To

Serial Monitor

1 (Arduino Uno R3)

SendClear

ULA 1 BIT

All changes saved

Simulator time: 00:02:53

CodeStop SimulationSend To

Serial Monitor

1 (Arduino Uno R3)

SendClear

ULA 1 BIT

All changes saved

Simulator time: 00:03:17

CodeStop SimulationSend To

Serial Monitor

1 (Arduino Uno R3)

SendClear

ULA 1 BIT

All changes saved

Simulator time: 00:03:43

CodeStop SimulationSend To

Serial Monitor

1 (Arduino Uno R3)

SendClear